Exploratory Data Analysis of Nobel Prize Laureates (1901–2025) using SQL

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In the context of global scientific and intellectual development, the Nobel Prize represents one of the most prestigious recognitions awarded to individuals and organizations for their contributions to humanity. The analysis of Nobel Prize data provides valuable insights into the distribution of knowledge production, gender representation, and the evolution of scientific fields over time.

This document presents an exploratory data analysis (EDA) of a dataset containing information about Nobel Prize laureates from 1901 to 2025. The analysis is conducted using SQL queries to examine patterns, trends, and key characteristics within the dataset.

The main objective of this report is to explore the structure of the dataset, analyze relevant variables, and extract meaningful insights through structured queries.

General Objective

To perform an exploratory data analysis of Nobel Prize laureates (1901–2025) using SQL in order to identify patterns, trends, and relevant insights.

Specific Objectives

- To examine the structure and variables of the dataset
- To analyze the distribution of Nobel Prizes by category
- To explore gender representation among laureates
- To identify trends over time
- To apply SQL queries for filtering, grouping, and aggregating data
- To interpret the results obtained from the analysis

Data Source

The dataset used in this analysis contains information on Nobel Prize laureates from 1901 to 2025. It includes variables such as year, category, gender, country of origin, affiliation, and prize details.

The data was loaded from a CSV file into a SQL environment for analysis.

Methodology

The analysis follows an exploratory data analysis (EDA) approach, which focuses on understanding the structure and characteristics of the dataset before applying more advanced techniques.

The process includes:

1. Loading the dataset into a SQL database
2. Inspecting the structure of the data
3. Formulating analytical questions
4. Executing SQL queries to answer those questions
5. Interpreting the results

Additionally, all SQL queries used in this project are available in a public GitHub repository, ensuring transparency and reproducibility of the analysis.

1.Initial exploration of the dataset

In order to understand the structure and organization of the data, an initial query was performed to display the first rows of the dataset. This step allows for the identification of available variables and provides a general overview of how the data is structured.

laureate_id	full_name	gender	birth_date	birth_city	birth_country	birth_country_now	death_date	death_city	death_country	year	category
160	Jacobus Henricus van 't Hoff	male	1852-08-30	Rotterdam	the Netherlands	the Netherlands	1911-03-01	Berlin	Germany	1901	Chemistry
569	Sully Prudhomme	male	1839-03-16	Paris	France	France	1907-09-07	Châtenay	France	1901	Literature
463	Frédéric Passy	male	1822-05-20	Paris	France	France	1912-06-12	Paris	France	1901	Peace
462	Jean Henry Dunant	male	1828-05-08	Geneva	Switzerland	Switzerland	1910-10-30	Heiden	Switzerland	1901	Peace
1	Wilhelm Conrad Röntgen	male	1845-03-27	Lennep	Prussia	Germany	1923-02-10	Munich	Germany	1901	Physics
293	Emil Adolf von Behring	male	1854-03-15	Hansdorf	Prussia	Poland	1917-03-31	Marburg	Germany	1901	Physiology or Medicine
161	Hermann Emil Fischer	male	1852-10-09	Euskirchen	Prussia	Germany	1919-07-15	Berlin	Germany	1902	Chemistry
571	Christian Matthias Theodor Mommsen	male	1817-11-30	Garding	Schleswig	Germany	1903-11-01	Charlottenburg	Germany	1902	Literature
465	Charles Albert Gobat	male	1843-05-21	Tramelan	Switzerland	Switzerland	1914-03-16	Bern	Switzerland	1902	Peace
464	Élie Ducommun	male	1833-02-19	Geneva	Switzerland	Switzerland	1906-12-07	Bern	Switzerland	1902	Peace

2. How many Nobel Prizes have been awarded by category?

category	total_awards
Physiology or Medicine	232
Physics	230
Chemistry	200
Peace	143
Literature	122
Economic Sciences	99

The results show the distribution of Nobel Prizes across different categories between 1901 and 2025. It is evident that Physiology or Medicine (232 awards), Physics (230 awards), and Chemistry (200 awards) are the categories with the highest number of prizes.

This pattern highlights the central role of scientific disciplines in the Nobel Prize framework, reflecting the historical importance of advancements in health, natural sciences, and technology for human development. These fields have consistently produced a high volume of recognized contributions due to their direct impact on improving quality of life and expanding scientific knowledge.

In contrast, categories such as Peace (143 awards) and Literature (122 awards) show a lower number of prizes. This difference may be explained by the more subjective nature of these fields, where recognition is often influenced by social, political, and cultural contexts rather than purely measurable outcomes.

Finally, Economic Sciences (99 awards) has the lowest number of prizes, which is consistent with the fact that this category was introduced later (in 1969). Therefore, it has had fewer years of participation compared to the original Nobel Prize categories.

Overall, the results suggest a strong emphasis on scientific achievements within the Nobel Prize system, while also reflecting the historical evolution and differing evaluation criteria across disciplines.

3. What is the gender distribution of Nobel laureates?

gender	total
male	927
female	68

The results reveal a significant gender imbalance in the distribution of Nobel Prize laureates. Out of the total, **927 awards have been granted to male recipients**, while only **68 have been awarded to female recipients**.

This stark difference highlights a historical underrepresentation of women in Nobel Prize recognition. The disparity can be associated with structural barriers that have limited women's access to education, research opportunities, and professional recognition, particularly in earlier decades.

Moreover, this gap is especially pronounced in scientific fields such as Physics, Chemistry, and Medicine, where female participation has historically been lower. Although progress has been made in recent years, the data suggests that gender inequality remains a persistent issue within the global academic and scientific community.

Overall, the findings reflect not only differences in recognition but also broader patterns of inequality in access to resources, opportunities, and visibility in high-impact fields.

4. How many prizes have been awarded over time?

year	total_awards
1901	6
1902	7
1903	7
1904	6
1905	5
1906	6
1907	6
1908	7
1909	7
1910	5
1911	6
1912	6
1913	5
1914	3
1915	4
1916	1

year	total_awards
2025	14
2024	12
2023	11
2022	14
2021	13
2020	12
2019	14
2018	13
2017	12
2016	11
2015	11
2014	13
2013	13
2012	10
2011	13
2010	11

The results show the number of Nobel Prizes awarded by year, comparing recent decades with the early years of the prize.

In recent years (2010–2025), the number of awards by year ranges approximately between 10 and 14 prizes, indicating a relatively stable and higher level of recognition. This reflects the expansion of scientific fields, increased collaboration, and the inclusion of more laureates sharing prizes.

In contrast, during the early years (1901–1916), the number of awards was significantly lower, generally ranging between 5 and 7 prizes per year, with some notable drops. For example, in 1914, 1915, and especially 1916 (only 1 award), there is a sharp decline.

This decrease can be associated with major historical events, particularly World War I, which disrupted scientific activity and international cooperation, leading to fewer awards being granted.

Overall, the comparison reveals a clear upward trend in the number of Nobel Prizes awarded over time. This growth reflects both the expansion of knowledge across disciplines and changes in the Nobel Prize system, such as the possibility of awarding prizes to multiple laureates for collaborative work.

In summary, the data suggests that the Nobel Prize has evolved from a more limited and irregular recognition system into a more consistent and inclusive global award structure.

5.Which countries have the most Nobel laureates?

birth_country_now	total_laureates
USA	299
United Kingdom	95
Germany	84
France	63
Sweden	30
Japan	30
Poland	29
Russia	29
Canada	22
the Netherlands	20

birth_country_now	total_laureates
Croatia	1
Myanmar	1
Costa Rica	1
Nigeria	1
Taiwan	1
Slovenia	1
Madagascar	1
Indonesia	1
Bulgaria	1
Guadeloupe, France	1

The results show a highly unequal distribution of Nobel Prize laureates across countries, with a strong concentration in a few nations and minimal representation in others.

On one hand, countries such as **the United States (299 laureates)**, **the United Kingdom (95)**, and **Germany (84)** dominate the ranking by a large margin. These nations have historically been global leaders in scientific research, higher education, and innovation. Their dominance can be attributed to factors such as strong academic institutions, high levels of research funding, and well-established scientific ecosystems.

Similarly, countries like **France (63)**, **Sweden (30)**, **Japan (30)**, **Poland (29)**, **Russia (29)**, **Canada (22)**, and **the Netherlands (20)** also show significant representation, reinforcing the idea that Nobel Prize recognition is concentrated in developed economies with robust research infrastructures.

On the other hand, several countries—including **Croatia, Myanmar, Costa Rica, Nigeria, Taiwan, Slovenia, Madagascar, Indonesia, and Bulgaria**—have only **one laureate each**.

This limited representation suggests disparities in access to scientific resources, funding, and global academic networks.

Overall, the data highlights a clear global inequality in the distribution of Nobel Prize recognition. It suggests that the production of high-impact knowledge and internationally recognized research is heavily concentrated in a small group of countries, while many regions remain underrepresented in the global scientific landscape.

6.Which categories have the highest female representation?

category	female_laureates
Peace	20
Literature	18
Physiology or Medicine	14
Chemistry	8
Physics	5
Economic Sciences	3

The results show how female Nobel laureates are distributed across different categories, revealing important differences between disciplines.

The categories with the highest number of female laureates are **Peace (20)** and **Literature (18)**. This suggests that women have had relatively greater recognition in fields that are more closely related to social impact, human rights, and cultural contributions.

In contrast, scientific fields show lower female representation. **Physiology or Medicine (14)** has a moderate presence of women, but **Chemistry (8)**, **Physics (5)**, and especially **Economic Sciences (3)** exhibit significantly lower numbers.

This pattern highlights a persistent gender gap in STEM disciplines (Science, Technology, Engineering, and Mathematics), where women have historically faced barriers to participation and recognition. The particularly low number in Physics and Economics suggests that these fields remain among the most unequal in terms of gender representation.

Overall, the results indicate that although female participation has increased over time, gender disparities are still evident, especially in scientific and technical fields, while more progress has been achieved in areas related to social sciences and humanities.

7.What is the distribution of prize shares by category?

category	prize_share	frequency
Chemistry	1	63
Chemistry	1/3	60
Chemistry	1/2	59
Chemistry	1/4	18
Economic Sciences	1/2	42
Economic Sciences	1/3	27
Economic Sciences	1	26
Economic Sciences	1/4	4
Literature	1	114
Literature	1/2	8
Peace	1	72
Peace	1/2	62
Peace	1/3	9
Physics	1/2	86
Physics	1/3	57
Physics	1	47
Physics	1/4	40
Physiology or Medicine	1/3	96
Physiology or Medicine	1/2	80
Physiology or Medicine	1	40
Physiology or Medicine	1/4	16

The distribution of prize shares across Nobel Prize categories highlights clear differences in the level of collaboration among disciplines. In **Chemistry**, the number of individual awards (63) is very close to those shared among three laureates (60) and two laureates (59), indicating a balanced mix between individual and collaborative work. Similarly, **Physics** shows a strong tendency toward collaboration, with most prizes shared between two (86) or three laureates (57), while individual awards (47) are less frequent. The relatively high number of prizes shared among four laureates (40) further reinforces the collaborative nature of this field.

In **Physiology or Medicine**, collaboration is even more pronounced, as the majority of prizes are shared among three laureates (96) and two laureates (80), compared to only 40 individual

awards. This suggests that research in this field often involves large teams and interdisciplinary efforts.

In contrast, **Literature** is overwhelmingly dominated by individual recognition, with 114 prizes awarded to a single laureate and only 8 shared prizes, reflecting the individual nature of literary achievements. The **Peace** category presents a more balanced pattern, with 72 individual awards and 62 shared between two laureates, indicating that both individual leadership and joint efforts are recognized.

Finally, in **Economic Sciences**, prizes are more commonly shared, particularly between two laureates (42), followed by three (27), while individual awards (26) are slightly less frequent. This suggests a moderate level of collaboration compared to other scientific fields.

Overall, the results demonstrate that scientific categories such as Physics, Chemistry, and Physiology or Medicine are highly collaborative, whereas non-scientific fields, especially Literature, tend to emphasize individual contributions

8.Which institutions are most frequently affiliated with Nobel laureates?

affiliation_name	total_laureates
University of California	39
Harvard University	28
Massachusetts Institute of Technology (MIT)	24
Stanford University	22
University of Chicago	20
Columbia University	18
Princeton University	18
California Institute of Technology (Caltech)	17
University of Cambridge	17
Rockefeller University	13

affiliation_name	total_laureates
Tufts University	1
Cold Spring Harbor Laboratory	1
The Wellcome Research Laboratories	1
Janelia Research Campus, Howard Hughes Medical ...	1
Université de Paris, Laboratoire Immuno-Hématolo...	1
Jackson Laboratory	1
Harvard University, Biological Laboratories	1
Central Research Laboratories, EMI	1
Glynn Research Laboratories	1
Toulouse School of Economics (TSE)	1

The results identify the institutions most frequently associated with Nobel Prize laureates, revealing a strong concentration in a small group of highly prestigious universities and research centers.

At the top of the list is the **University of California (39 laureates)**, followed by **Harvard University (28)**, **MIT (24)**, and **Stanford University (22)**. These institutions are globally recognized for their excellence in research, innovation, and academic influence. Their leading positions reflect their ability to attract top talent, secure significant funding, and maintain strong research infrastructures.

Other prominent institutions include the **University of Chicago (20)**, **Columbia University (18)**, **Princeton University (18)**, **Caltech (17)**, and the **University of Cambridge (17)**. These universities have historically played a central role in scientific advancement and continue to be key contributors to high-impact research.

On the other hand, the second table shows several institutions with only **one affiliated laureate**, such as Tufts University, Cold Spring Harbor Laboratory, and Toulouse School of Economics. This indicates that while many institutions contribute to knowledge production, only a few consistently produce Nobel-level research.

Overall, the results highlight a clear concentration of Nobel laureates in elite institutions, particularly in the United States and Europe. This suggests that access to advanced research environments, funding, and academic networks plays a crucial role in achieving high-level scientific recognition.

9. Which categories have grown the most over time?

category	total
Physics	71
Chemistry	68
Physiology or Medicine	63
Economic Sciences	55
Peace	37
Literature	26

The results show the distribution of Nobel Prizes by category in recent years, allowing us to identify which fields have experienced the most growth over time.

The categories with the highest number of awards are **Physics (71)**, **Chemistry (68)**, and **Physiology or Medicine (63)**. This indicates that scientific disciplines continue to dominate the Nobel Prize landscape and have shown sustained growth over time. These fields are

closely linked to technological advancement, innovation, and global challenges, which explains their continued expansion and recognition.

Economic Sciences (55) also shows significant growth, especially considering that it is a relatively newer category compared to the traditional Nobel Prize fields. Its high number of awards suggests an increasing relevance of economics in understanding global systems and public policy.

In contrast, **Peace (37)** and **Literature (26)** have lower counts, indicating more limited growth. This may be due to the more selective and subjective nature of these categories, where awards are often influenced by social, political, and cultural factors rather than continuous measurable advancements.

Overall, the results suggest that Nobel Prize growth is strongly concentrated in scientific and technical fields, while other areas such as humanities and social impact have grown at a slower pace. This reflects broader global priorities in research, innovation, and problem-solving.

10.Who has won more than one Nobel Prize?

full_name	total_prizes
Marie Curie, née Skłodowska	2
Linus Carl Pauling	2
John Bardeen	2
Frederick Sanger	2
K. Barry Sharpless	2

The data shows that Marie Curie, Linus Carl Pauling, John Bardeen, Frederick Sanger, and K. Barry Sharpless are distinguished by having received the Nobel Prize twice. This highlights a small group of exceptionally influential scientists whose contributions have had a lasting and significant impact across different fields. For instance, Marie Curie was awarded for her pioneering research in both Physics and Chemistry, while Linus Pauling uniquely received prizes in Chemistry and Peace, reflecting both scientific and humanitarian achievements. Similarly, John Bardeen is notable for winning twice in Physics, and both Frederick Sanger

and K. Barry Sharpless were recognized twice in Chemistry. Overall, this pattern suggests that repeated Nobel recognition is rare and typically associated with groundbreaking, transformative work that advances knowledge across disciplines or over extended periods of time.

11.Who are the most recent winners of the Nobel Prize in Economics?

full_name	year	category
Philippe Aghion	2025	Economic Sciences
Peter Howitt	2025	Economic Sciences
Joel Mokyr	2025	Economic Sciences
Daron Acemoglu	2024	Economic Sciences
James A. Robinson	2024	Economic Sciences
Simon Henry Roberts Johnson	2024	Economic Sciences
Claudia Goldin	2023	Economic Sciences
Ben S. Bernanke	2022	Economic Sciences
Douglas W. Diamond	2022	Economic Sciences
Philip H. Dybvig	2022	Economic Sciences

The most recent Nobel Prize winners in Economic Sciences from 2022 to 2025 reflect key contributions to understanding how economies grow, function, and respond to challenges. **In 2025, Philippe Aghion, Peter Howitt, and Joel Mokyr** were awarded for their work on economic growth theory, particularly emphasizing the role of innovation, technological progress, and knowledge in driving long-term economic development.

In 2024, Daron Acemoglu, James A. Robinson, and Simon Johnson received the prize for their research on the role of institutions in economic development, demonstrating how political and economic systems—such as governance, laws, and democracy—directly influence a country's prosperity.

In 2023, Claudia Goldin was recognized for her pioneering research on women in the labor market, especially her analysis of the gender wage gap and the historical factors shaping female workforce participation.

In 2022, Ben S. Bernanke, Douglas W. Diamond, and Philip H. Dybvig were awarded for their studies on banks and financial crises, explaining the critical role of banks in the economy, the causes of financial instability, and how crises can be mitigated. Overall, these awards highlight the importance of research in areas such as innovation, institutional development, labor economics, and financial stability, all of which are essential for understanding modern economic systems.

Conclusions

This exploratory data analysis of Nobel Prize laureates (1901–2025) using SQL revealed important patterns related to scientific recognition, gender representation, institutional concentration, and global inequalities in knowledge production. The results show that scientific categories such as Physics, Chemistry, and Physiology or Medicine dominate the Nobel Prize system, reflecting the global importance of innovation, research, and technological advancement.

The analysis also highlighted a strong concentration of Nobel laureates in a small number of countries and elite institutions, particularly in the United States and Europe. This suggests that access to high-quality research environments, funding opportunities, and academic networks plays a significant role in achieving international recognition.

Regarding gender representation, the findings demonstrate that women remain underrepresented among Nobel laureates, especially in STEM (science, technology, engineering, and mathematics). However, the data also shows gradual progress over time, particularly in areas such as Peace and Literature, where female participation has become more visible. Rather than focusing on limited access to education, the results can be interpreted as evidence of broader historical and structural inequalities in professional recognition and participation within scientific and academic spaces.

Additionally, the project revealed how collaboration has become increasingly important in scientific research, as many prizes in Physics, Chemistry, and Medicine are shared among multiple laureates. This reflects the growing complexity and interdisciplinary nature of modern scientific discoveries.

References and data sources

1. Dataset

Kaggle. (2026). *Nobel Prize Laureates (1901–2025)*.
Disponible en:
https://www.kaggle.com/datasets/elvisbui/nobel-prize-laureates-1901-2025

2. Project Repository

Camargo Vega, L. (2026). *Exploratory Data Analysis of Nobel Prize Laureates (1901–2025) using SQL*
Repository available on GitHub:
https://github.com/LuisaCamargoVega/Exploratory-Data-Analysis-of-Nobel-Prize-Laureates-1901-2025-using-SQL

3. Tool used

RunSQL. Platform used for executing SQL queries during analysis.